

ECON 633

Mathematics for Economists

Department of Economics
McGill University
Summer 2022

Instructor: Julia Koh

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Regular classes: Daily from August 15th to August 30th, 10.00am - 1.00pm, LEA 424.

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Course Overview and Learning Outcomes

The course is aimed at bringing students' knowledge of calculus, analysis, linear algebra and optimization theory up to levels required by the graduate macroeconomics and microeconomics courses and other courses.

Prerequisite

One variable calculus and linear algebra. (These will be only briefly reviewed.)

Course Materials

The lectures will be based on material from the following textbooks.

“Mathematics for Economics”, by Hoy et al (*HOY*)

“Mathematics for Economists”, by Simon and Blume (*SB*)

Assignments

Daily assignment: will be provided in the lecture notes. The assignments need to be written by LaTeX (LaTeX will be introduced in the beginning of the class.)

(*The lecture notes will be provided after the class.)

Course Content

1. Review of univariate calculus and linear algebra (*SB: Ch 7 - 11*)
 - (a) Foundations and applications of univariate calculus
 - (b) Matrix algebra
 - (c) Linear independence
2. Multivariate calculus (*SB: Ch 12 - 15*)
 - (a) Sets and functions
 - (b) Multivariate calculus
 - (c) Implicit functions and derivatives
3. Optimization (*SB: Ch 16 - 22*)
 - (a) Unconstrained optimization
 - (b) Constrained optimization
 - (c) Homogeneous and homothetic functions
4. Eigenvalues and Eigenvectors (*SB: Ch 23*)
5. Programming language in Economics: MATLAB

Course Schedule

- August 15th: introduction + LaTeX.
- August 16th - 17th: Review of univariate calculus and linear algebra.
- August 18th - 19th: Sets and review of real analysis.
- August 22nd - 23rd: Multivariate calculus.
- August 24th - 24th: Optimization.
- August 25th: Homogenous & Homothetic functions.
- August 26th: Convex Analysis.

- August 29th: Eigenvalues and Eigenvectors.
- August 30th: MATLAB.

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